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Cognitive Psychology

Extensive Phenomenological Overlap Between Training-Induced and Naturally-Occurring Synaesthetic Experiences

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Synaesthesia is a condition defined by additional perceptual experiences, which are automatically and consistently triggered by specific inducing stimuli. The associative nature of synaesthesia has motivated attempts to induce synaesthesia by means of associative learning. Two recent studies of this kind highlighted the potential for perceptual plasticity even in adulthood, by demonstrating that extensive associative training can generate not only behavioural and neurophysiological markers of synaesthesia, but also synaesthesia-like phenomenology. However, while the results of these studies provided tantalising evidence that a learning component may be involved in the development of synesthetic phenomenology, they only provided superficial descriptions regarding the training-related changes in induced synaesthesia-like (*Induced*) experience. Therefore, it was not possible to assess how closely the phenomenology of *Induced* and naturally occurring grapheme-colour synaesthesia (*Lifelong*) overlap. Here, we addressed this question by providing a new qualitative analysis, using *grounded theory*, of the phenomenological changes associated with learning new perceptual phenomenology (*Induced* group) and comparing the descriptive similarities in colour experience to equivalent qualitative data acquired from a new group of *Lifelong* participants. Using this approach, we were able to directly compare associated colour experiences between the *Induced* and *Lifelong* group to assess how closely these two types of novel perceptual experience align. Our results reveal that induced and synaesthetic experience are remarkably similar, displaying a high degree of phenomenological overlap across multiple experiential categories, including: stability of experience, location of colour experience, shape of co-occurring colour experience, relative strength of colour experience and automaticity of colour experience. Our results exemplify the benefits of qualitative methods by providing new evidence that intensive training of letter-colour associations can alter conscious perceptual experiences in non-synaesthetes, and that such alterations produce synaesthesia-like phenomenology, which substantially resembles experiences described in natural grapheme-colour synaesthesia. Our results have implications for the plasticity of visual perception and the role of learning and development in establishing perceptual traits.

1.0. Introduction

Synaesthesia is defined by the presence of additional perceptual experiences, which are automatically and consistently triggered by specific inducing stimuli. For example, in grapheme-colour synaesthesia, the presence of grapheme (letter) stimuli triggers additional (concurrent) experiences of colour – even when the inducing stimulus is itself monochromatic. The associative nature of grapheme-colour synaesthesia has led researchers to posit the involvement of a learning component in the development of synaesthetic associations (for review see Rothen and Meier (2014)). This has prompted a number of studies to investigate whether it is possible to train non-synaesthetic individuals to develop synaesthesia-like experiences (Arend et al., 2022; Bor et al., 2014; Colizoli et al., 2012, 2016, 2017; Meier & Rothen, 2009; Rothen et al., 2011, 2018). Two previous studies of this kind revealed dramatic plasticity in

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human visual perception by showing that the majority of participants in the first study (Bor et al., 2014), and all participants in the second study (Rothen et al., 2018) reported phenomenology suggestive of grapheme-colour synaesthesia. However, these studies only provided superficial descriptions of induced synaesthesia-like (*Induced*) changes in phenomenology. Therefore, it was not possible to establish how closely the perceptual phenomenology of *Induced* and naturally occurring synaesthesia (*Lifelong*) aligned. To investigate the precise phenomenological overlap between *Induced* and *Lifelong* colour experience, we performed a new in-depth qualitative analysis of the transcripts of the semi-structured interview obtained following the completion of the training protocol (data taken from Rothen, Schwartzman, et al. (2018), interviews were not recorded in Bor et al. (2014)). We then assessed the descriptive similarities between the *Induced* group and equivalent qualitative data from a new sample of *Lifelong* participants, using methodologies from qualitative phenomenology, allowing us to directly compare induced and synaesthetic subjective reports to identical questions about their colour experiences.

The defining hallmarks of *Lifelong* synaesthesia remains the subject of debate, mainly due to the heterogeneity of synaesthetic experience. However, some of the widely agreed upon hallmarks of *Lifelong* synaesthesia include the *automaticity* of the concurrent experience (i.e., not experienced as being under voluntary control) (Dixon et al., 2000; Meier & Rothen, 2009; Ward et al., 2007), and the *consistency* of the associations (i.e., repeated presentations of an inducer will elicit highly similar concurrent experiences) (Baron-Cohen et al., 1996; Eagleman et al., 2007; Rothen, Seth, et al., 2013). Another well-established feature of *Lifelong* synaesthesia relates to differences in the perceived location of synaesthetic experience. Some grapheme-colour synaesthetes, referred to as *projectors*, experience their concurrent outside bodily space, somehow 'projected' into the world. Others, referred to as *associators*, report that their experiences exist within their internal mental space without any prominent spatiality (Dixon et al., 2004; Rothen, Tsakanikos, et al., 2013; Simner, 2013).

Motivated by the apparent associative nature of synaesthesia, two previous synaesthesia training studies from our lab (Bor et al., 2014; Rothen et al., 2018) used an associative training regime that was designed to cement 13 specific letter-colour associations in an attempt to simulate the possible real-life development of synaesthesia far more closely than achieved by previous training studies (for more detail of the training regime used, see Rothen et al. (2018)). The regime repeatedly paired 'inducer' and 'concurrent' stimuli across adaptive memory and reading tasks. Both studies found that overtraining letter-colour associations resulted in a profile of behavioural performance similar to those seen in genuine synaesthesia. Using the gold-standard consistency test (Eagleman et al., 2007; Rothen, Seth, et al., 2013), both studies reported that consistency performance for trained letter-colour pairs passed the threshold indicative of synaesthetic experience. In addition, both studies assessed the automaticity of synesthetic associations using a synaesthetic equivalent of the Stroop test, finding that participants displayed evidence of automatic synaesthesia-like behaviour for trained compared to untrained letters (demonstrated by greater interference effects and slower response times in incongruent trials [see (Meier & Rothen, 2009; Ward et al., 2007)]). Critically, both studies were also successful in generating striking subjective reports of synaesthesia-like phenomenology. However, while these studies reported representative quotes and subjective reports that described the automaticity of trained synaesthetic phenomenology, the superficial level of detail meant that it was not possible to establish how closely training induced alterations in perceptual phenomenology matched the concurrent experiences of genuine grapheme-colour synaesthetes.

In this study, we investigate the precise phenomenological overlap between *Induced* and *Lifelong* synaesthetic experience by comparing the results of a new extended qualitative analysis (see section 2.3) of the phenomenological changes associated with learning new perceptual phenomenology (*Induced* group) with a comparable analysis of new interview data acquired from a group of grapheme-colour synaesthetes (*Lifelong* group). Comparing subjective reports to identical questions about colour experiences we found substantial phenomenological overlap across multiple experiential categories between *Induced* and *Lifelong* groups. Our extensive first-persons descriptions – of both *Induced* and *Lifelong* synaesthesia - extend previous reports of the phenomenological parallels between training induced and natural synaesthetic experience. The detailed qualitative assessment of such overlap is important not only for understanding synaesthesia, but for shedding light on the potential for perceptual plasticity in general, and the roles of development and learning in establishing perceptual traits.

2.0. Material and Methods

2.1. Participants

The *Induced* group consisted of 18 non-synaesthetes (15 women, mean age = 23, SD = 3.08), whose data and interview recordings were taken from Rothen et al. (2018). In this previous study we only reported simple summary metrics obtained from these recordings (Rothen et al., 2018). The *Lifelong* group consisted of a new sample of 15 grapheme-colour synaesthetes (14 women, mean age = 43.3, SD = 11.43). Results of the grapheme-colour consistency test revealed that all *Lifelong* participants were below the threshold indicative of *Lifelong* grapheme-colour synaesthesia (M = 76.53; SD = 21). Studies were undertaken with the understanding and written consent of each *Induced* participant. All *Lifelong* participants were recruited from the University of Sussex synaesthesia database based on indicative consistency scores of grapheme-colour synaesthesia (www.synesthete.org; Eagleman et al. (2007)). Informed consent was obtained from *Lifelong* participants prior to the beginning of the interview. Both studies were approved by the University of Sussex ethics committee.

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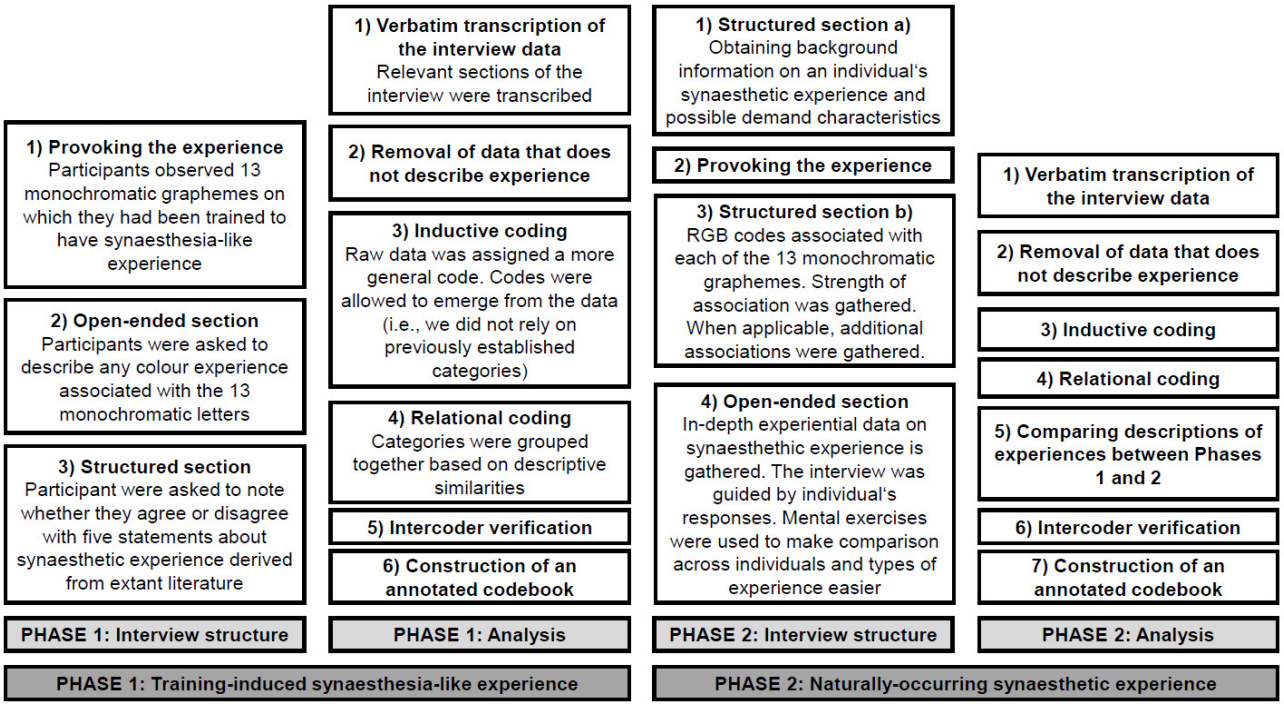


Figure 1. Schema outlining the structure of the multi-phase research used in this study.

2.2. Behavioural tests

All participants completed the internet-based standardized grapheme-colour consistency test (Eagleman et al., 2007). The test presents each participant with the graphemes A–Z three times in randomized order. For each grapheme, participants select the colour that best matches their putative concurrent or synaesthesia-like experience. The *Induced* group performed this test twice, once before and once upon completion of the training paradigm, to assess the effect of the training on the consistency of the trained associations (for details see Rothen et al., 2018). As part of their inclusion in the University of Sussex's synaesthesia database, all *Lifelong* participants completed the grapheme-colour consistency test (once).

2.3. Phenomenological analysis

Our research design was constructed around three related approaches in empirical phenomenology and qualitative research: multi-phase research (Kordeš & Klauser, 2016), sequential analysis (Flick, 2009) and theoretical sampling (Charmaz, 2006) (see Figure 1). Our use of the term 'phase' is derived from the notion of multi-phase research in qualitative studies, here we use two Phases of research, phase 1 related to the *Induced* group and phase 2 the *Lifelong* group. To summarise this approach, to begin with the acquisition of qualitative data is open-ended (i.e., interviews are less structured) and the logic of analysis is inductive (i.e., sections of raw data are assigned descriptive tags based on the content of the data itself, rather than a previously established theoretical framework). In each subsequent phase, as the object of inquiry becomes more defined, the acquisition of qualitative data becomes more focused (moving towards semi-structured and structured interviews, and the logic of the analysis is inductive-deductive; i.e., earlier insights are used to guide the analysis, while additional categories are constructed based on novel findings) (Flick, 2009; Kordeš & Klauser, 2016). In the final phases, the logic of analysis becomes deductive, and all the data are subjected to the same analysis frameworks.

While most qualitative phenomenological frameworks (Larkin et al., 2021; Stanghellini & Mancini, 2017) are based on situating a given aspect of lived experience in the broader context of a person's life, our approach aimed at identifying specific aspects of cognition that are more or less isolatable from the participants' broader existential situation. Thus, in our re-analysis of the data, we wished to move away from constructive an interpretative narrative about synaesthesia but identifying objectively verifiable properties of synesthetic experience.

Based on this approach, in the first phase we analysed transcripts from the semi-structured interview used in Rothen et al. (2018) to closely examine the properties of colour experience in the *Induced* group, with the aim of discerning its main descriptive properties (see Figure 1). The semi-structured interview enquired about the core characteristics of synaesthesia (e.g., consistency, automaticity, unidirectionality etc.), based on empirical evidence from previous investigations of *Lifelong*.

The second phase was designed using the *Induced* participants' subjective reports to narrow the focus of enquiry towards producing more precise and focused qualitative data from *Lifelong* participants (see Figure 1). The new semi-structured interview was constructed so that it acquired data conducive to comparison with the *Induced* participants, while also allowing for further insights about *Lifelong* experience. Following identification of all experiential cat-

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egories of *Lifelong* experience, we returned to the *Induced* data and constructed a single, unified analytical framework (see the *annotated codebook*, in supplemental material), allowing for a direct comparison of the descriptive similarities of colour experiences between the two groups. The order of the Phases was based on the historical dependencies between the two studies. In the following subsections, we present the protocol for each Phase of research.

2.4. Phase 1: Induced Group. Subjective reports of induced synaesthesia-like experiences

In the first phase, following the completion of a 5-week training battery, all (*Induced*) participants performed an in-person semi-structured interview designed to assess their perceptual phenomenology during exposure to 13 monochromatic trained letters [for details see Rothen et al. (2018), and for a summary see Figure 1]. Only sections of the interview that explicitly asked participants to describe their colour experiences were transcribed. These sections included responses to the question "Look at this page that has the 13 letters you have been trained over the last 5 weeks, to associate with 13 specific colours. I want you to describe any associated colour experience you have when looking at these letters". Due to the nature of open-ended interviews, some queries emerged from the reports participants volunteered. Within the interview, this led to a subset of 14 out of 18 participants being asked to compare the vividness of their strongest trained synaesthesia-like colour experience (e.g., 'r' is red) to visual imagery associated with a life-long colour association of a real-world object (e.g., the specific shade of red associated with an English post-box) using the question:

"I want you to bring to mind a typical everyday colour association from the real world that you have developed over the course of your life, such as *post-boxes are red* or the *sky is blue*. Can you give me an example.... You know the specific colour associated with this real-world object. How is the experience of imagining the colour associated with this real-world object in your 'mind's-eye similar or different to you 'seeing' your synaesthetic associations?"

Next, we constrained the focus of responses in a funnel-like structure frequently used in semi-structured interviews (Minichiello & Aroni, 1990), by including responses and discussion to a single forced choice question that required participants to localise their trained synaesthesia-like experiences in space. The forced choice question was: "Which statement characterises your grapheme-colour associations best?" Participants were asked to select one of the following responses that most accurately reflected their experience:

"Whenever I see a letter...

- There is only that letter, but no colour at all. I can't even think of an associated colour, no matter how hard I try.
- I know the associated colour, but I never see it. (Petitmengin, 2009)
- I see the colour outside my head (i.e., a few inches away).
- I see the colour floating on the surface wherever the letter is."

The interviews were transcribed in two stages following established methods of analysing subjective reports (Petitmengin, 2009; Valenzuela-Moguillansky & Vásquez-Rosati, 2019). First, the description of relevant subjective experiences were highlighted, while other descriptions (such as clarification about the question, or irrelevant discussion) were removed.

Second, transcripts for all participants (N = 18) (transcripts are available https://osf.io/e367d/?view_only=dd61d42daa7a4c848023b89bd38789f8) were analysed according to the principles of *content analysis*, whereby codes are not assigned to the data based on the concrete words used, but rather according to the underlying meaning (Braun et al., 2012). We used *inductive coding*, meaning that we ascribed abstract, more general descriptions to the raw data without recourse to established theoretical constructs (cf. Charmaz (2006)). This approach recognises that the knowledge gathered through interviews is jointly constructed by the researcher and participant, rather than being discovered in an observer-independent fashion (Mills et al., 2006). After inductive coding, we grouped specific codes together based on their structural (i.e., descriptive) similarities (so-called *relational coding;* Flick, 2009).

The validity of data acquisition and analysis was evaluated using two methods: a) *the annotated codebook* and b) *intercoder verification*. The annotated codebook approach (Nelson, 2017) serves three purposes. Firstly, it organises qualitative data by establishing meaningful relationships between individual reports. Secondly, a fully specified codebook (i.e., a codebook in which meaningful relationships between all the categories is established) establishes that the gathered data is 'deep' enough for theory-construction (Fusch & Ness, 2015; O'reilly & Parker, 2013). Thirdly, it describes the coding process in sufficient detail to enable independent replication.

The annotated codebook consists of a *saturation grid* and the codebook itself. The saturation grid is a tabulation in which for each participant we note new instances of codes established from the transcripts. Once we observe no new categories for an individual, conceptual depth has been achieved (also called saturation), and it is no longer necessary to recruit further participants (Kerr et al., 2010). Table 1 displays a condensed version of the saturation grid for the *Induced* group (a fully specified saturation grid for the *Induced* group can be found in the supplementary materials). As can be seen, conceptual depth was reached with participant 17.

The codebook was constructed according to standard approaches in empirical phenomenology (e.g., (Brod et al., 2009; Hurlburt & Heavey, 2006; Kordes et al., 2019)). Each entry in the codebook is specified according to the following elements: name, description, subcategories, examples, and considerations. Considerations describe any specific differences between similar categories as well as concerns that may have emerged during the coding process (see supplementary materials for the full annotated code-

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Table 1. Condensed version of a sample saturation grid for the Induced group.

Induced Participant #	1	2	3	4	5	6	7	9	11	12	13	14	16	17	18	20	21	22
New codes	9	3	1	2	0	1	1	0	0	0	1	0	0	1	0	0	0	0

books https://osf.io/e367d/?view_only=dd61d42daa7a4c848023b89bd38789f8).

The validity of the first analysis phase was also addressed using intercoder verification. Two coders (authors DS and AO) coded all transcripts independently and then compared their results for consistency (De Wet & Erasmus, 2005). Experiential categories were only identified and carried forward for analysis if identified by both coders.

2.5. Phase 2: Lifelong Group. Subjective reports of naturally-occurring synaesthetic experiences

In the second phase, we collected additional data using an updated semi-structured interview from individuals with *Lifelong* (all but one interview were performed via remote video call). A potential concern with *Lifelong* data was that the remote video-based context of the interview may have affected their subjective reports. Reassuringly, responses to the question 'Do your synaesthetic experiences change depending on whether you are reading a text from a computer screen or a piece of paper?', revealed that the synaesthetic experiences of all participants (N=15) were not affected by the medium on which the grapheme was presented.

The interview was modified using findings from the first phase (*Induced*). The interview began by guiding *Lifelong* participants towards observing and reporting on the main phenomenological dimensions that were extracted from the first analysis phase (*Induced*). Using the logic of inductive-deductive analysis, the narrowing of focus allowed us to refine our understanding of the specific aspects of *Lifelong* experience, while also allowing for emergence of novel categories. While the interview structure remained similar in both Phases, Phase 2 acquired more in-depth subjective reports, thereby allowing for comparison with the *Induced* group. Phenomenological methodology is based on "bracketing" the taken-for granted assumptions about conscious experience and providing descriptions of what was concretely present in experience in a given moment. In this way, whenever participants in Phase 2 referred to existing concepts (e.g., our own pre-established categories, well-known ideas from the sciences of the mind or pop psychology), we inquired in more detail to obtain longer descriptions of their experience. In this way, equivalence between categories was established on descriptive similarity of experiential reports, rather than merely single words or phrases used.

The semi-structured interview followed a tripartite structure. First, we obtained general information about participants' synaesthetic experiences, including questions relevant to the situational demands of the interview (see supplemental material for interview questions). We also collected demographic information and we noted whether participants were professionally involved with academic mind-science disciplines (e.g., psychology, neuroscience). These participants tended to provide theoretically-laden description of experience and therefore required additional attention during the interview to retain focus on theory-independent properties of their perceptual experience.

In the second part of the semi-structured interview, using an identical procedure to Rothen et al. (2018), *Lifelong* participants viewed 13 monochromatic graphemes, which were presented (via screen-share), and they were asked to report on their associated colour experience. Following the presentation of each grapheme, they were asked to verbally identify the RGB code that best reflected their specific colour experience in relation to the grapheme, using an online colour picker (https://htmlcolorcodes.com/color-picker/). Participants also indicated the strength of the association between the grapheme and the colour on a scale from 1 (weakest imaginable) to 10 (strongest imaginable). Finally, participants indicated the automaticity of their synaesthetic experience on a scale from 1 (fully wilful) to 10 (fully automatic).

Participants were then asked a number of open-ended questions and the same mental exercise and forced choice question as with the *Induced* group. The open-ended questions were:

- Can you describe the details of the colour experience (shape, letter, block etc.; the precise location of your synaesthetic experience).
- When looking at a letter, for instance, 'r', are you aware of both the actual colour of the letter on the page (black) and your synaesthetic colour experience? Or does one overlay the other?
- Can you describe how your synaesthetic colour experience are similar/not similar to your perception of the colour of a real-world object?
- Do your synaesthetic experiences change if the grapheme is capitalized or in a different font?
- Are there any other types of experiences you have had in your life, which are similar to your synaesthetic experiences?

The audio recordings of the interviews were transcribed verbatim. Whenever participants referred to colour shades that were unfamiliar to the researcher by name, a picture of that colour was included in the transcript. If the participants referred to phonemes rather than graphemes, the audio in question was transcribed using the International Phonetic Alphabet. Subsequently, the transcripts from the *Lifelong* and *Induced* groups were analysed using the same codebook to afford direct comparison (i.e., deductive coding was used). We constructed a taxonomy of experiential categories that allowed for comparison of phenomenological dimensions of *Induced* with *Lifelong* experiences.

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Table 2. Condensed version of saturation grid for the Lifelong group.

Lifelong Participant #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
New codes	11	10	2	1	1	3	2	0	1	1	0	0	0	0	0

The validity of the second-phase (*Lifelong*) data was established using the same methods as for the first phase, i.e., through the construction of an annotated codebook and intercoder verification (DS and AO). As with the *Induced* group, we determined conceptual depth by constructing a saturation grid (Table 2), which revealed that we had reached saturation by participant 10.

A chi-square test of independence was performed to examine the relation between *Induced* and *Lifelong* groups for each experiential category. We only included commensurate values in the analysis (i.e., as the data from the *Lifelong* group was more fine-grained than the *Induced* group, not all experiential categories existed in both groups). We excluded instances where no values could be induced from the raw data; due to participants not reporting on a specific category. This statistical comparison for each experiential category should be viewed as additional information to aid in the interpretation of qualitative findings, rather than a rigorous quantitative analysis.

3.0. Results

In the following, we label participants from the *Induced* group as (*Induced*-x) for participant x, and similarly for the *Lifelong* group (*Lifelong*-x).

3.1. Comparisons of experiential categories between Induced and Lifelong groups

We now examine each experiential category, their relative similarities and differences between *Induced* and *Lifelong* groups, and the continuity between these reports and those found in the wider synaesthesia literature. The extent to which the present findings mirror findings from this literature helps establish the validity of the present analysis (*theoretical sampling,* Charmaz, 2006).

3.1.1. Stability of colour experience

This category reflects a high-level description of how varied the colour experience for each grapheme was for each participant across all presented graphemes. It primarily encompasses the vividness of the associated colour experience and how this differs depending on the situation. Although other experiential categories, such as *location*, *shape* and *automaticity,* may also be evaluated with regards to their overall stability, these categories were here identified as separate dimensions of experience.

As is common in *Lifelong* synaesthesia (Ward, 2013), the majority of *Induced* participants reported that their experience was *heterogeneous* (15 out of 17), displaying a high degree of variability in the vividness of their synaesthesia-like colour experience associated with each letter. For example, the letter 'r' in general produced a more vivid colour experience (red) than the letter 'u' (grey). Only two *Induced* participants reported a *homogenous* colour experience for all graphemes (see Table 4).

We found a similar degree of variability in the vividness of concurrent experience for *Lifelong* participants. Ten *Lifelong* participants (N = 15) reported that the vividness of their concurrent experiences was highly varied between individual graphemes (coded as *heterogeneous*), with only 5 participants reporting that their experiences were homogenous across all graphemes (see Table 4). An exploratory chi-square test of independence found no significant difference between the *Induced* and *Lifelong* groups in terms of the stability of their colour experience: $\chi 2(1, N = 32) = 1.091$, $p = 0.296$ (Cramér's V measure of association, φc, was 0.185).

Examining if *Lifelong* experiences had remained consistent throughout their lifetimes (specific to the *Lifelong* group), we found that eight participants reported that their experiences had remained stable. Four participants reported minor changes throughout their lives, including changes in the synaesthetic colour associated with single letters or words (e.g., *Lifelong*-02 reports the word for *Saturday* shifted from blue to grey at the onset of adulthood), or variability in the intensity of synaesthetic colours (e.g., both *Lifelong*-11 and *Lifelong*-04 report that they sometimes worry that they may have "lost their colours"). Three participants reported more significant changes, *Lifelong*-05 and *Lifelong*-11, for example, described that all of their synaesthetic colours had become faded with age (cf. also Meier, Rothen, and Walter (2014)).

> *Lifelong*-05: If anything, it got a little weaker from when I was younger [...] I just, I sometimes stop and think about what colour is a letter or some letters are not a strong colour anymore and I'm pretty sure they used to be.

The second dimension of stability refers to whether the colour of a person's synaesthetic experience changes with context. This dimension is specific to the *Lifelong* group – All synaesthetic phenomenology for the *Induced* group is, by definition, dependent on the context of the study. Eight *Lifelong* participants (N = 15) reported the presence of concurrent colours even during goal-oriented behaviour, such as reading. Conversely, seven participants reported that when reading, they were able to push their synaesthetic experiences out of their awareness.

3.1.2. Location of colour experience

The higher-order level of description (*location*) within this category describes whether the participants colour experience was located within their mental space or if it was

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externally localised. The lower-order level (*location* (*specified*)) describes the location of the colour experience in relation to the grapheme: *letter-adjacent* denotes situations in which the colour was experienced proximal to the letter, while *letter-overlapping* describes the associated colour experience occurring directly over the letter. This category of experience was observed in both the *Induced* and *Lifelong* groups.

The *Induced* group showed a similar variation in *location*, as has been reported for *Lifelong* synaesthesia (Dixon et al., 2004; Simner, 2013). Some (9 out of 17 *Induced* participants) reported their synaesthesia-like experience as occurring in external space (*projector-like*), for example:

> *Induced*-04: I see the same letter with the colour, I'm looking at 'o' right now and I can see the orange 'o' floating around right now.

Whereas others (8 out of 17) reported them as occurring within their "mind's eye" (*associator-like*), for example:

> *Induced*-13: I don't actually see these letters on the page change colour; I just read 'g' and see my whole head full of green.

Responses to the forced choice question: "Which statement characterises your grapheme-colour associations best?" revealed a slightly different, but generally congruent, picture (see Table 3). Twelve out of 18 *Induced*[1] participants reported that their synaesthesia-like experience occurred in front of their "mind's eye", and the rest reported that they occurred 'floating on the surface wherever the letter is'.

Within the *Lifelong* group, we found fewer projector-like descriptions (3 out of 15), for example:

> Participant *Lifelong*-15: As I look at any of the words or letters, even if I see them, you know, I can see that they are black or white, I can also see their (synaesthetic) colour.

However, the majority (12 out of 15) described their concurrent experience as occurring within their mental space (see Table 4), for example:

> Participant *Lifelong*-03: It's [sighs] it's certainly not something that is out there in the world, it certainly not something that I could reach out and touch. It is definitely [pause] 100% all internal.

This finding was supported by responses to the forced choice question (see Table 3). As responses to the forced choice question were taken in the context of a face-to-face interview, *Lifelong* participants had the opportunity to elaborate on their answers. The table therefore contains two counts: *primary* and *secondary* responses for individuals with *Lifelong*. The primary response represents the count of their typical synaesthetic experience, whereas the secondary response represents a count of their infrequent synaesthetic experiences (usually qualified in the interviews with phrases "It happens sometimes," or "It happens in specific circumstances").

An exploratory chi-square test of independence, found no significant difference between the *Induced* and *Lifelong* groups for the experiential category *location*: $\chi 2(1, N = 32) = 2.418$, $p = 0.120$ (Cramér's V measure of association, φc, was 0.275).

We also observed a large degree of heterogeneity for both groups when describing the precise location of their colour experiences (*location (specified)*). For the *Induced* group, where this information was available, approximately half (7 out of 13) described their colour experience as overlapping with the grapheme, while the others (6 out of 13) described it as occurring near or around the grapheme.

Within the *Lifelong* group we observed two cases of *letter-adjacent* colour experience, and five cases of *letter-overlapping* colour experiences (N = 7 total). However, for the remaining *Lifelong* participants, we identified an additional experiential category that was not present in the *Induced* group, in which the location of concurrent experience could not be discerned, because the colour experiences were tied to vague descriptions that occurred in the participant's mental space. These participants described their synaesthetic experience as containing a component (N = 6) or consisted solely (N = 1) of propositional knowledge of the grapheme-colour association. In other words, the participants *knew* the concurrent colours without being able to localise them in space (Rothen, Tsakanikos, et al., 2013; Ward et al., 2007). Finally, one participant (*Lifelong*-04), reported experiencing an 'embodied connection' to the concurrent colour, such that she experienced her concurrent through her body. An exploratory chi-square test of independence found no significant difference between the *Induced* and *Lifelong* groups for the experiential category *location* (*specified*): $\chi 2(1, N = 21) = 0.257$, $p = 0.612$ (Cramér's V measure of association, φc, was 0.111).

On first inspection, reports from both groups appear to support the associator/projector subtype distinction. However, a more detailed examination of reports from the *Lifelong* group reveal a more complicated picture in which four aspects of reported experience challenge this distinction. First, unlike classic descriptions (Ward et al., 2007), we observed a number of individuals in which both associator and projector phenomenology co-occurred (8 out of 15) (for a similar finding, see (Edquist et al., 2006)). Consider the following example:

> Participant *Lifelong*-14: Well, it is in my head, but it is also where I see it (on the page). So, it's a difficult one.

Participant *Lifelong*-03, reports that when external stimuli appear spontaneously or take him by surprise, his

1 The discrepancy in the number of participants stems from the fact that *Induced*-21 did not report synaesthetic phenomenology in the open-ended portion of the interview. He nonetheless offered a response to the forced-choice question.

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Table 3. Absolute frequencies of forced-choice responses to the question "Which statement characterises your grapheme-colour associations best?" for Induced and Lifelong groups.

Forced choice statement	*Induced:* Responses	*Lifelong:* Primary response	*Lifelong:* Secondary response
Whenever I see a letter, there is only that letter, but no colour at all. I can't even think of an associated colour, no matter how hard I try.	0	0	0
Whenever I see a letter, I know the associated colour, but I never see it.	0	2	2
Whenever I see a letter, I see the colour in front of my mind's eye.	12	11	0
Whenever I see a letter, I see the colour outside my head (i.e., a few inches away).	0	0	2
Whenever I see a letter, I see the colour floating on the surface wherever the letter is.	6	2	4

synaesthetic experience was described as occurring in the *outside world,* but when elicited in the artificially constrained environment of the interview, his concurrent experience was described as residing within his mental space.

Second, some *Lifelong* participants (7 out of 15) described *knowing* the associated concurrent colour without being able to localise it in space. Others have described this instantiation of grapheme-colour synaesthesia as 'know-associator', that is, such synaesthetes simply know the colour associated with each letter but do not have a visual representation of it (Edquist et al., 2006; Ward et al., 2007).

Third, within cases that described their grapheme-colour associations as propositional knowledge, five out of the six *Lifelong* participants described their concurrent phenomenology as consisting of multiple streams of experience. Rather than being aware of a concrete colour, the participants experienced a vague awareness or a feeling of a colour.

Finally, one participant (*Lifelong*-04), reported experiencing an embodied connection to the concurrent colour. Her description implies an active participation in attending to the colour such that she experiences her concurrent through her body (matching what has been referred to as *attentional dispositions*, (Kordes et al., 2019) and *existential orientation* in phenomenological literature (Petitmengin, 2009; Ratcliffe, 2008); and *overarching states of mind* in neuroscientific literature (Herz et al., 2020). Again, this description falls outside of the classical subtype classification.

Together, the experiential categories *location* and *location* (*specified*) reveal a large degree of overlap between *Induced* and *Lifelong* experiences. We found that *Induced* and *Lifelong* experiences were described as occurring either in external space (*projector*), or within their "mind's eye" (*associator*)(location). For the experiential category location (specified), we found similar descriptions of the specific location of the colour experience in both groups, with the majority describing the concurrent experiences as occurring either adjacent to, or overlapping with the letter. Within the *Lifelong* group, we discovered an additional category of experience, 'know-associator', in which the location of concurrent experience could not be discerned, as they only were aware of the colour associated with each letter but did not have a visual representation of the colour.

3.1.3. Shape of colour experience

This category describes differences in the visual quality of the colour experience. All of the *Induced* and the majority of the *Lifelong* group described the same three forms of colour experience for this category, reporting that either their colour experience mirrored the shape of the grapheme (*Induced*: 7 out of 14; *Lifelong*: 8 out of 15) (coded as *letter*), that their associated colour experience had a *discernible geometric shape* (e.g., a block of colour) (*Induced*: 3 out of 14; *Lifelong*: 1 out of 15), or that their colour experience exhibited a shapeless presence of colour or an *aura* (*Induced*: 4 out of 14; *Lifelong*: 6 out of 15) – in one extreme case (*Lifelong*-04), the participants' entire mental space was filled with the awareness of the associated colour; (category *totality of colour*; see Table 4).

In terms of the specific shape of synaesthetic colour experience, we identified an additional experiential category within the *Lifelong* group. There were some cases where colour experiences were explicitly described as having no shape (*none;* N = 4).

Some values for the category *shape* may overlap; i.e., an individual may associate one letter with a specific shape and another letter with a different shape. Thus, we observed one case where the shape of the letter co-occurred with the shape of an aura, and one case where it co-occurred with totality of colour.

An exploratory chi-square test of independence found no significant difference between the *Induced* and *Lifelong* groups for this experiential category: $\chi 2(2, N = 22) = 1.228$, $p = 0.541$ (Cramér's V measure of association, φc, was 0.236).

3.1.4. Relative strength of colour experience

We constructed three values to describe the relative strength of synaesthesia-like experience in comparison to visual imagery associated with a life-long colour association of a real-world object: *weaker than non-synaesthetic colour association, stronger than non-synaesthetic colour association,* and *equal to non-synaesthetic colour association.*

We found strong descriptive similarities between both groups for this experiential category, with the majority of participants reporting that their associated synaesthetic/synaesthesia-like colour experience was equal to or weaker than visual imagery associated with a non-synaesthetic

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colour association (*Induced*: 11 out of 16; *Lifelong*: 11 out of 15). Consider the following representative *Induced* example of this position:

> Participant *Induced*-21: OK, er, on a scale of 1- 10 if I gave paper is white as 10 as the strongest, I would put 'r' somewhere in the 7-8 range.

Five *Induced* and two *Lifelong* participants reported that their synaesthetic experience was stronger than visual imagery associated with a life-long colour association (see Table 4). For two their mental imagery was stronger than their synaesthetic experience, and for two *Lifelong* participants, the experience of mental imagery and grapheme-colour synaesthesia were too different to be comparable. Interestingly, nine *Lifelong* participants reported that there was no qualitative difference between their synaesthetic experience and visual imagery of a real-world colour association, for example:

> Researcher: *Bananas are yellow* is okay. Compare that colour association with *red is rusty brown,* or sorry, *R is rusty brown.* [...] How are these two statements similar or different in your experience?
> Participant *Lifelong*-10: They are pretty much exactly the same, to be honest. [pause] There's no difference.

3.1.5. Automaticity of colour experience

Automaticity of colour experience is a category that refers to whether participants (in either the *Induced* or the *Lifelong* group) experience synaesthetic concurrents as something that happens to them (i.e., an automatic, passive event), or something that they do (i.e., a wilful mental action). In Rothen et al. (2018), in response to a forced choice question 12 out of 18 *Induced* participants reported that their concurrent experience was automatic, 3 were undecided and 3 said the translation required wilful effort. For example:

> Participant *Induced*-03: Yeah, I think it is automatic, as I was scanning down, each colour came to me pretty instantly. I kinda see it in my mind as a replication of the letter but in its colour, ...

In contrast, the present qualitative analysis found that for the majority of Induced participants (9 out of 14), synaesthesia-like experience required mental effort to occur, with only five reporting that the experience was automatic. For example, consider the following representative response about the automaticity of the participant's colour experience:

> Participant *Induced*-16: No, I think I have to look at them each individually to sort of, um, yeah I'd have to look at them each individually to picture the colour, but it comes quite easily.

In contrast to the *Induced* responses, we found that the majority of *Lifelong* participants (8 out of 15) described their synaesthetic colour experience as fully automatic (see Table 4), which was confirmed using a forced choice question, in which participants rated the average automaticity of their synaesthetic experience from 1-10, 10 being fully automatic; the average score across all *Lifelong* participants was 8.99 (N = 15, SD = 1.62). Only one *Lifelong* participant described their synaesthetic colour experiences as being *wilful*.

While the *Induced* group only reported on the distinction between *wilful* and *automatic* experiences, we identified three additional aspects of *Lifelong* experience for this category. The first is *contextually varied experience*, which describes individuals whose synaesthetic experiences displayed different levels of automaticity based on their immediate situation. For example, as previously mentioned, three *Lifelong* participants reported that while reading prose they were able to shift their attention such that they were not aware of their synaesthetic experiences.

The second additional experiential category is *semi-automatic experience,* which refers to situations in which participants experience the option of bringing their concurrent experience to the forefront of their awareness, or to not pay attention to it at all. Two *Lifelong* participants reported *semi-automatic* experiences. Consider the following report:

> Participant *Lifelong*-04: I don't focus in on the colour quite as much as if I am focusing on it. So, to say, like, for example, I am reading a novel. I don't focus on the colours that are there. So, they become sort of pushed back out of my conscious thought until I individually think of a letter.... It's a background noise type thing.

The third additional experiential category is *reflective association.* This category describes reports of a grapheme being able to enter into awareness without an associated colour experience. The colour experience only becomes present when they consider or mindfully reflect on the grapheme. If they do not assume this reflective stance, no synaesthetic experience occurs. Surprisingly, similar to reports from the *Induced* group that their colour experience required mental effort, we found codes referring to the *reflective* nature of concurrent experiences in the majority of *Lifelong* participants (10 out of 15).

The category *reflective association* contains two values: *phase-in* and *phase-out*. The former describes situations in which participants perform a mental gesture in order to bring the synaesthetic experience to the forefront of their awareness, as in the following example:

> Participant *Lifelong*-05: I don't *see* the colour when I look at the word on the page. But when I envision a letter in my mind, it has a colour. Words in my mind have a colour, and certainly days of the week and numbers very strongly have a colour. But if I look at your list [of letters], I don't see the colours when I see the letters.

Conversely, the value *phase-out* refers to the mental gesture whereby participants move their synaesthetic experience to the back of their mind (see excerpt from participant *Lifelong*-04, above). Similar reports of mental action altering perceptual experience have also been described in other forms of synaesthesia, such as sequence-space synaesthesia. In a single case study, the concurrent experience was described as occurring within a "mental room", with the participant reporting the ability to selectively shift their at-

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Table 4. Frequency distribution of phenomenological dimensions associated with Induced and Lifelong experience.

Phenomenological dimensions of *Induced* and *Lifelong* experiences							
Stability of experience	Value	Heterogeneous	Homogenous	NA			
	f(*Induced*)	15	2	1			
	f(*Lifelong*)	10	5	0			
Strength of association	Value	Weaker than colour association	Equal to colour association	Stronger than colour association	NA		
	f(*Induced*)	6	5	5	5		
	f(*Lifelong*)	3	5	2	2		
Location	Value	Outside location	Mental space	NA			
	f(*Induced*)	9	8	1			
	f(*Lifelong*)	3	12	0			
Location (specified)	Value	Letter-adjacent	Letter-overlapping	NA			
	f(*Induced*)	6	7	5			
	f(*Lifelong*)	2	6	5			
Shape	Value	Letter	Discernible shape	Totality of colour	Aura	None	NA
	f(*Induced*)	7	3	2	2	/	4
	f(*Lifelong*)	8	1	1	3	4	0
Automaticity	Value	Willful	Automatic	Varied	Semi-automatic	NA	
	f(*Induced*)	10	5	/	/	3	
	f(*Lifelong*)	1	8	3	2	0	

The symbol "/" represents values that were not induced from the data in that specific group.

tention to either the mental room in which the concurrent occurred, or to the real world (Gould et al., 2014).

The results of an exploratory chi-square test of independence identified a significant difference between the *Induced* and *Lifelong* groups for the experiential category *automaticity*: $\chi 2(1, N = 24) = 4.934$, $p = .026$ (Cramér's V measure of association, φc, was 0.453). (Note, as mentioned earlier, the automaticity of letter-colour associations varied from grapheme to grapheme).

Table 4 summarises the frequency distribution of all phenomenological dimensions between the *Induced* and *Lifelong* groups.

4.0. Discussion

Here, we report an extended qualitative analysis that investigated the phenomenological similarities between induced synaesthesia-like (*Induced*) and naturally occurring synaesthetic (*Lifelong*) experience. Our extensive first-person descriptions – of both *Induced* and *Lifelong* experiences - reveal a substantial phenomenological overlap between *Induced* and *Lifelong* experience, extending previous reports of the parallels between these two types of novel perceptual experience (Bor et al., 2014; Rothen et al., 2018). In addition to the clear descriptive similarities in reports describing their respective colour experiences, we identified a number of specific experiential categories that were shared between the two groups, including: stability of experience, location of colour experience, shape of co-occurring colour experience, relative strength of colour experience and automaticity of colour experience.

While previous research has been successful in establishing superficial similarities in phenomenology between training induced and natural synaesthesia (Bor et al., 2014; Colizoli et al., 2012, 2016; Meier & Rothen, 2009; Rothen et al., 2011), the present analysis goes much further and deeper. Our results reveal that overtraining letter-colour associations results in a phenomenological profile that bear striking similarities across multiple experiential categories to *Lifelong* grapheme-colour synaesthesia. For both groups, colour experience was heterogeneous, displaying a high degree of variability in the vividness of colour experience associated with each letter. Both groups described their colour experiences as occurring along the associator/projector continuum, either occurring in external space (projector), or within their "mind's eye" (associator). Both groups described the same three specific forms of colour experience, that either mirrored the shape of the grapheme, exhibited a discernible geometric shape, or was described as a shapeless aura. The majority of both groups described their colour experiences as occurring either adjacent to or overlapping with the letter. Finally, the relative strength of

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their colour experience was reported by the majority of both groups as being equal to or weaker than visual imagery associated with a non-synaesthetic colour association. While we refrain from ascribing a phenomenological equivalence between *Induced* and *Lifelong* synaesthetic experiences, in combination with our previous findings (Bor et al., 2014; Rothen et al., 2018), our results provide strong evidence that the new perceptual phenomenology, which occurred as the result of intensive and adaptive training of letter-colour associations produced a phenomenological profile that bear marked similarities to those found in *Lifelong* synaesthesia.

Although synaesthesia was originally considered to occur due to a genetic predisposition (Asher et al., 2009), the close resemblance in perceptual phenomenology between *Induced* and *Lifelong* synaesthetic experience argues against the idea that synaesthetic phenomenology can only occur in a rare subset of the population that exhibit such a genetic predisposition (Asher et al., 2009; Barnett et al., 2008; Witthoft & Winawer, 2006). Instead, our results further support the notion that Lifelong synaesthesia may include a significant developmental component. Even in cases where a genetic predisposition may lead to the development of synaesthesia, the observation that in most forms of synaesthesia concurrent experiences are triggered by cultural artefacts, suggests that the development of natural synaesthesia must include a substantial dependence on learning and prior experience (Bor et al., 2014; Rothen et al., 2018; Rothen & Meier, 2014; Witthoft & Winawer, 2006).

We now briefly consider the implications of our findings for synaesthesia research. Natural synaesthetes have often been characterised along a projector to associator continuum (Dixon et al., 2000, 2004). In the present study, both *Induced* and *Lifelong* groups reported that their synaesthetic experience occurred either within their mind's eye or in external space. However, within the *Lifelong* group we found a number of aspects of experience that falls outside of the classical subtype classification. For example, we found that some *Lifelong* participants reported both types of description, while others reported that there was no specific spatiality associated with their concurrent experience. Others have described this instantiation of grapheme-colour synaesthesia as 'know-associator', that is, such synaesthetes simply know the colour associated with each letter but do not have a visual representation of it (Edquist et al., 2006; Ward et al., 2007). This non-perceptual subtype, does not fit neatly within the projector/association classification (even though these participants display consistency for letter-colour pairings), and may reflect a weak form of synaesthesia in which all perceptual aspects of a concurrent experience have faded over time. Our findings therefore suggest the need for a more finessed characterisation of synaesthetic phenomenology than provided by spatial location of concurrent experiences (Edquist et al., 2006; Simner, 2013; Ward et al., 2007).

Examining the relative strength of synaesthetic experience within the *Lifelong* group revealed that for the majority their concurrent experience was equal to or weaker than visual imagery associated with a non-synaesthetic colour association, suggesting that for some there is a level of equivalence between visual imagery and their synaesthetic experience. Due to the hidden nature of synaesthetic experience, ever since it was first described, debate has continued over whether synaesthetic experiences should be viewed as distinct from other forms of private subjective experience, such as vivid mental imagery (Albright, 2012; Deroy & Spence, 2013; Lupyan et al., 2023; Price, 2009).

Synaesthetic experiences described as occurring within the "mind's eye" evidently share similarities with descriptions of visual mental imagery. By contrast, projector variants of grapheme-colour synaesthesia are evidently dissimilar from imagery, and the automaticity characteristic of much synaesthetic experience is also distinct from the wilful, voluntary nature of most imagery. However, we found that some *Lifelong* participants reported volitional control over their synaesthetic experience, while the majority reported that their concurrent experience only occurred following a volitional *reflective association*. We acknowledge that visual imagery is a complex and flexible phenomenon; that likely consists of many components (Kosslyn et al., 1984). However, these reports suggests that for some synaesthetes their concurrent experience exhibits a degree of volitional control, which they report as equivalent to visual imagery of a real-world colour association (Bergmann et al., 2016). Further work is warranted to examine if individual variations in phenomenal, perceptual differences, or cognitive characteristics - such as visual mental imagery - affects dimensions of grapheme-colour synaesthesia experience, such as the vividness of concurrent colour experience. In one ongoing study addressing this topic, we and others are measuring 'perceptual diversity' in many different dimensions of perception and cognition and across tens of thousands of people worldwide. The study is called The Perception Census (https://perceptioncensus.dreamachine.world/).

When examining the automaticity of colour experience in both groups, our results initially suggested that *Lifelong* experiences were typically perceived as occurring automatically, whereas *Induced* experiences were generally reported as being wilful. However, results from the category *reflective association* reveal a more complex picture: for *Lifelong* participants, while the closed-form questions point to concurrent experiences being automatic, responses from the category *reflective association* suggests that for the majority of participants they are experienced as being wilful. We posit that this apparent contradiction is best explained by a common finding in contemporary phenomenology. Namely, under careful observation, many of our experiences seem to be underlined by a variety of micro-gestures; that is, mental acts that have become so habitual as to be seen as automatic under gross observation (Petitmengin, 2021). When asked about the automaticity of their experience, similar to the *Induced* group, the different prompts (open-ended vs. closed-form) may have caused the *Lifelong* participants to provide an answer based on different aspects of their experience (Heimann, 2020; Nagel, 2014; Sikka et al., 2017). The open-ended questions (within the context of the interview) may have been interpreted by the participants as enquiring

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about the awareness of colour in their visual consciousness, whereas the closed-form measures may have been interpreted as referring to their propositional knowledge about the grapheme-colour association (e.g., for the majority of *Lifelong* participants the associative link between 'R' and 'red' was automatic in a participant's experience, whereas the visual experience of redness elicited by the letter 'R' appeared to be associated with the reflection on the letter itself). This possibility was investigated in a case study of an individual with multiple forms of synaesthesia (Oblak et al., 2021). The study suggests that the link between inducer and concurrent is neither immediate nor passive. Rather, it amounts to a process of attending to the inducer, wherein the concurrent gradually evolves from amodal knowing to modally present experience.

It is worth emphasizing that the phenomenological descriptions reported here are by no means exhaustive, particularly with regards to *Lifelong* synaesthesia (*Lifelong*). Indeed, even within the present set of transcripts, additional experiential themes and categories were identified, which remain to be fully explored.

Prominent among these additional categories is the notion of veridicality or 'perceptual presence'. Seth (2014) has argued that most synaesthetic concurrents lack veridicality or presence, in the sense that they are not experienced as being 'real' properties of the world. In the present study, the majority of both groups provided similar descriptions about the veridicality of their concurrent experiences, which were compatible with a lack of perceptual presence. For example, in one example, participant *Induced*-12 describes their experience as follows: "I would say that they feel roughly the same, but it might sound stupid, but like tomato is more of a real colour, but when I think of the green (trained colour) I think of it on a computer screen and it's a sort of not a real thing, so it doesn't feel as natural. When you think of a tomato you think about how the light hits something, that shape, and that's part of the colour". However, unlike properties such as spatial location, we did not observe systematic variation in descriptions of perceptual presence in our data. Further research is needed to examine the nature and variability of perceptual presence in both *Induced* and *Lifelong* groups, as well as the cognitive and neural underpinnings that explain this aspect of experience.

Considering the potential factors that may have influenced the results of this study, we note that demographic information from the *Lifelong* group revealed that six out of the fifteen participants worked in mind-science areas (e.g., psychology, neuroscience), raising the possibility that they may have provided theoretically-laden descriptions of their experience. *Lifelong* synaesthesia represents a relatively small subset of the population (Simner et al., 2006), with many researchers relying on the same group of synaesthetes across multiple studies. Future studies should take care to make sure that participants' detailed knowledge of synaesthesia is not influencing their results, creating a circular confirmation of the condition (the so-called *looping effect,* Sperber et al., 1995); that is, to avoid entrenching a situation in which the profile of synaesthesia described in the literature occurs precisely because participants are aware of normative responses associated with the condition.

The possibility that implicit demand characteristics may drive experience (as well as shape behavioural and neurophysiological responses) cannot be excluded in this study, especially for subjects who score high on scales of hypnotic suggestibility or 'phenomenological control' (Lush et al., 2020). Phenomenological control refers to an individual's ability to alter what they experience, both within and outside of the hypnotic context, in ways that are consistent with their plans and goals (Dienes et al., 2020; Lush et al., 2020). Recent work has investigated the possible influence of phenomenological control in generating experiences in mirror-sensory synaesthesias: mirror touch, in which the observation of someone being touched elicits a reported tactile sensation in the observer; and vicarious pain, in which observed pain elicits reports of experienced pain (Lush et al., 2020). Lush et al. (2020) found that hypnotisability scores strongly predicted mirror-sensory synaesthesia responses, suggesting that some of the experienced touch and pain sensations may have been the result of the participants' capacity to 'create' the experiences of touch via phenomenological control. The authors suggest that in the case of mirror-sensory synaesthesias, individuals may habitually (but involuntarily) implement phenomenological control in everyday life, when it is in-line with their goals, creating tactile sensations. These findings raise the possibility that habitual phenomenological control may also underlie the reported colour experiences in both *Induced* and *Lifelong* grapheme-colour synaesthesia. To address this question, future studies investigating synaesthetic phenomenology should also utilize scales such as the Phenomenological Control Scale (PCS), to examine possible relations to individual differences in phenomenological control ability (Lush et al., 2021). Such habitual phenomenological control may be deployed to aid in the acquisition of specific skills, such as learning to read, which may account for the high prevalence of grapheme-colour synaesthesia.

Extensions of the approach applied here may be used to advance neurophenomenological accounts of synaesthesia. For example, first-person data about the vividness of synaesthetic colour experience associated with each letter could be used as a heuristic to describe and quantify the neural dynamics and correlates associated with discrete synaesthetic experiences. Indeed, previous studies have already shown how individual differences in synaesthetic phenomenology (localisation) are associated with characteristic neural responses (Gould van Praag et al., 2016). It remains an open question to what extent associative training paradigms that lead to dramatic alterations in phenomenology, such as those described here, can modify structural and large-scale dynamical features of the human brain.

The phenomenological dimensions identified here display a large degree of overlap with previous reports of synaesthesia (e.g., Dixon et al. (2000); Simner (2013); Ward (2013)). It is therefore likely that these dimensions are typ-

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ical of a wider grapheme-colour synaesthesia population. Indeed, our saturation analysis (while not exhaustive) supports this assumption, by showing that conceptual depth was reached in both groups. However, we acknowledge the conclusions drawn in our study are limited to our participants and the training regime used (Rothen et al., 2018). Further work is required to examine if the close phenomenological overlap between *Induced* and *Lifelong* synaesthesia found here apply to larger and more diverse populations.

5.0. Conclusion

Our results demonstrate that intensive training of letter-colour associations can alter conscious perceptual experiences in non-synaesthetes, and that such alterations produce a profile of synaesthesia-like phenomenology, which substantially resembles experiences described in *Lifelong* grapheme-colour synaesthesia. The dramatic changes in perceptual phenomenology, and their phenomenological similarities with *Lifelong* synaesthesia, highlights the likely importance of experience and learning in the development of *Lifelong* synaesthesia and human visual perception more generally.

Author Contributions

DJS and AO designed the study. DJS and AO collected interview data and contributed to analysis and interpretation of data. DJS and AO drafted the manuscript. All authors provided critical revisions to the manuscript.

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Competing Interests

The authors declare no competing interests.

Supplemental Material

Study materials including the annotated codebook and transcripts of all interviews can be found on this paper's project page on the OSF:

https://osf.io/e367d/?view_only=dd61d42daa7a4c848023b89bd38789f8

Data Accessibility Statement

All materials and participant data used in this study can be found on this paper's project page on the OSF:

https://osf.io/e367d/?view_only=dd61d42daa7a4c848023b89bd38789f8

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